

The Three Domains: Scouting, Tracking, and Strategy

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Sport Analytics Begins With a Decision

For any sport analytics problem, ask:

1. Who will use the analysis?
2. What decision are they trying to make?
3. What information could improve that decision?
4. What are the consequences of being wrong?
5. What data might provide that information?

Overview

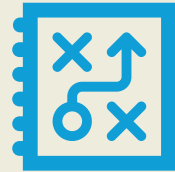
- Understand where sport performance analytics is applied and how each domain supports real-world decisions.



**Scouting & Talent
Evaluation**



Athlete Tracking



**Strategy & Tactical
Optimization**

Domains and the Different Decision Environments

- **Scouting and Talent Evaluation**

- Evaluating athletes, projecting future performance, and informing recruitment.

- **Athlete Tracking and Monitoring**

- Understanding training, competition demands, and athlete responses over time.

- **Strategy and Tactical Analysis**

- Understanding how games unfold and supporting decisions about how to compete.

The domains describe **where** analytics is applied and **which decisions** it supports.

Domain 1: Scouting & Talent Evaluation

- Evaluate current ability, understand development, and estimate future fit or performance.

Common decisions

- Recruit or draft an athlete
- Compare potential prospects
- Identify development priorities
- Build a balanced roster

Common information

- Performance history
- Training history
- Physical and technical profile
- Competition context
- Age

Evaluation Depends on the Reference Point

Consider the claim: “This athlete is an excellent shooter.”

- Excellent compared with what?
 - Teammates?
 - Players at the same position?
 - Players of the same age?
 - Players in the current league?
 - Players at the level being considered?

A number becomes evaluative only when it has a meaningful comparison.

Evidence From Different Sources

In scouting:

- **Statistical performance:** what the athlete has produced
- **Video and observation:** how that performance occurred
- **Physical testing:** the athlete's specific capabilities
- **Interviews and background:** the athlete's role, experience, and environment

Domain 2: Athlete Tracking and Monitoring

- Understand what athletes do and how they respond across training and competition.

Common decisions

- Adjust training
- Plan recovery
- Prepare for competition demands
- Evaluate return-to-play progression

Common information

- External load
- Internal load
- Wellness and readiness
- Performance-test results

Tracking Connects the Training Plan, Completed Work, and Athlete Response

- What was prescribed?
 - The planned training or competition demand
- What did the athlete complete?
 - Duration, distance, accelerations, impacts, repetitions
- How did the athlete respond?
 - Heart rate, RPE, soreness, wellness, performance changes

Workload has meaning only when interpreted in context.

Monitoring Data Requires Interpretation

- Athlete monitoring data can identify unusual patterns or changes.
- A flag indicates that something may deserve **attention** but does not diagnose a problem.
- Coaches and practitioners must interpret the information within the athlete's broader **context**.

Domain 3: Strategy & Tactical Analysis

- Understand patterns of play and support decisions about how to compete.

Common information

- Events and outcomes
- Spatial patterns
- Situational tendencies
- Probabilities and expected value

Common decisions

- Select lineups or rotations
- Prepare for an opponent
- Choose offensive or defensive tactics
- Make in-game adjustments

Situational Context and Game Event

- The value of a shot, pass, or play depends on:
 - **When** it occurred: score, time remaining, and game situation
 - **Where** it occurred: player location and defensive pressure
 - **Who** was involved: opponent, matchup, and teammates on the court or field

Strategy analysis asks not only what happened, but when, where, and against whom.

Developing an Analytical Question

Analytical Questions Serve Different Purposes

- Within any domain, we might want to:
 - **Describe** what occurred
 - **Diagnose** why it may have occurred
 - **Predict** what may occur next
 - **Prescribe** what action should be taken

Four Types of Analytical Questions



1. Descriptive

What happened?



2. Diagnostic

Why / how did it happen?



3. Predictive

What is likely to happen?



4. Prescriptive

What should we do about it?

These are different purposes. **Not** a ranking

Descriptive Analytics

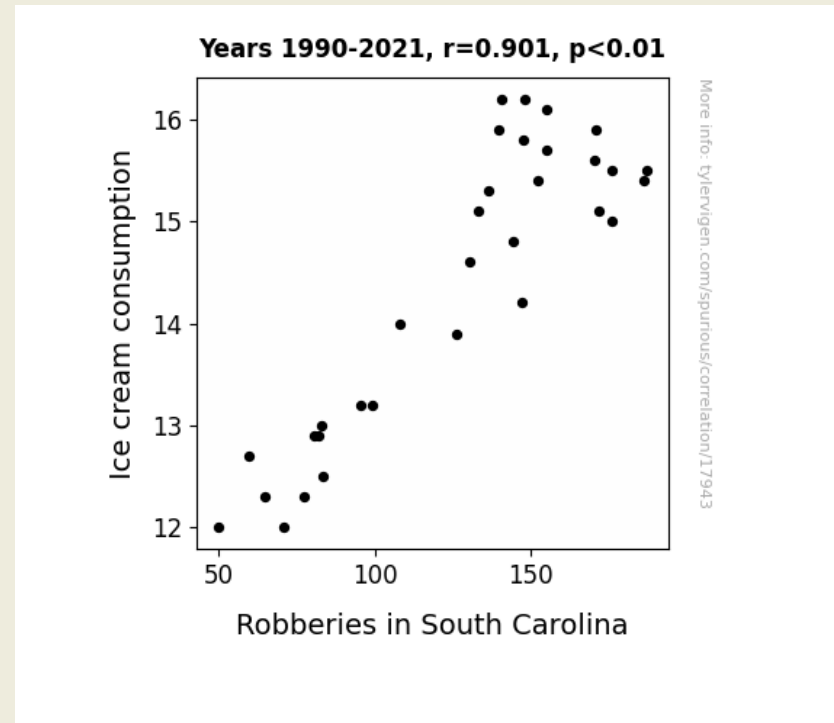
- Descriptive analytics summarizes past or current data to show what occurred.
- Identifies patterns and results, but it does not explain why they occurred.
- Examples:
 - **Tracking:** How much high-intensity work did the athlete complete?
 - **Scouting:** What are the prospect's current performance statistics?
 - **Strategy:** Which players created the most scoring opportunities?

Diagnostic Analytics

- Diagnostic analytics uses data to investigate the **possible** reasons behind an outcome.
- May identify **possible** explanations, but an observed relationship **does not prove** causation.
- Examples:
 - Did workload increase because practice was longer or more intense?
 - Did the player perform differently against stronger opponents?
 - Did turnovers contribute to the team allowing more shots?

Spurious Correlation in Diagnostic Analytics

- Relationship in which two or more events or variables are associated but not causally related
- Relationship may be driven by coincidence or a **confounding variable**



Predictive Analytics

- Predictive analytics uses available information to estimate an unknown outcome, often in the future.
- Predictions are uncertain and depend on the quality of the data and assumptions.
- Examples:
 - How is the athlete likely to perform in the next game?
 - Which prospect is most likely to succeed at the next level?
 - What is the probability that a possession produces a score?

Prescriptive Analytics

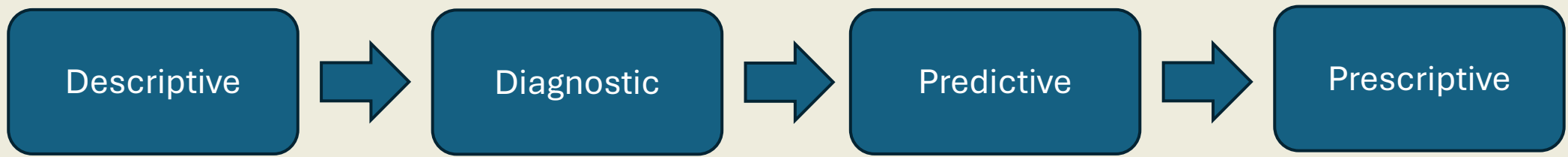
- Prescriptive analytics uses evidence to compare possible actions and recommend what should be done.
- May **guide** decisions, but the final choice also depends on context, priorities, experience, and human judgment.
- Examples:
 - Should the athlete's training be modified?
 - Which prospect should the team select?
 - Which defensive strategy should the team use?

Analytics Chain in Each Domain

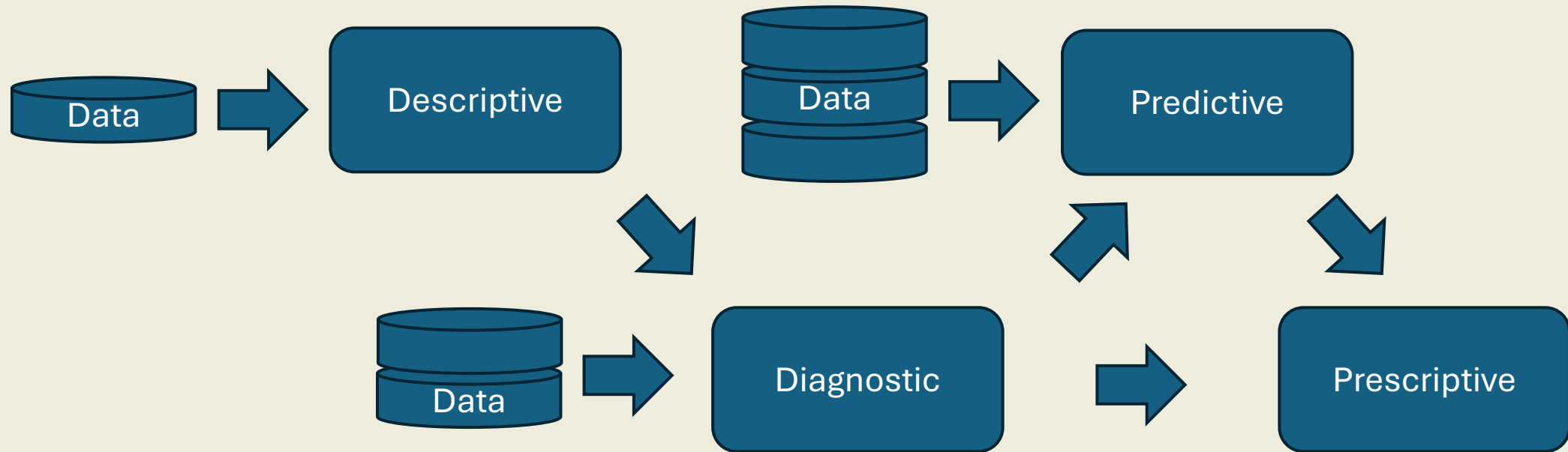
Domain	Descriptive	Diagnostic	Predictive	Prescriptive
Scouting	What has the player done?	Why has performance varied?	How might the player perform at the next level?	Should we recruit or draft the player?
Athlete monitoring	What work did the athlete complete?	Why did workload or readiness change?	How might the athlete respond tomorrow?	Should training be modified?
Game strategy	What happened during the game?	Why did the strategy succeed or fail?	What is likely against the next opponent?	What strategy should the team use?

Analytics Is a Connected Process

- Description can support diagnosis, diagnosis can inform prediction, and all three can contribute to a recommendation.



Real Analytical Work Is Rarely Linear



More Complex ≠ More Valuable

- Much of the daily work in sport analytics involves:
 - Collecting, cleaning, and organizing data
 - Defining metrics consistently
 - Identifying meaningful changes and patterns
 - Communicating findings to decision-makers

Predictive models can be valuable, but only when they address a real decision and are supported by reliable foundational work.

Summary & Looking Ahead

- The three domains describe where sport analytics is applied.
- Descriptive, diagnostic, predictive, and prescriptive questions describe what the analysis is intended to accomplish.
- Next: Turning practitioner concerns into clear, testable analytical questions